

Project: Fundamental Booster

Objective:

Create an Interactive Personal Data Collector application in Python that captures, processes, and displays personal information from the user using fundamental Python functions and concepts. This project should demonstrate your understanding of `print()` and `input()` functions, data types, variables, operators, type casting, and built-in functions like `id()` and `type()`.

Requirements:

Detailed Use of `print()` and `input()` Functions:

- Use `input()` to gather various types of personal information from the user, such as name, age, height, and favourite number.
- Use `print()` to display formatted information and messages to guide the user through each step.

Data Types, Variables & Operators:

- Store each piece of collected information in a variable with an appropriate data type.
- Demonstrate the use of arithmetic operators (e.g., `+`, `-`, `*`, `/`) to perform calculations like determining the user's birth year based on their age.
- Use string concatenation or formatted strings to construct user-friendly output messages.

Type Casting Constructors:

- Prompt the user for data, and ensure appropriate type casting (e.g., casting age to an integer and height to a float).
- Convert data to other types as necessary and explain type conversion clearly in the output (e.g., converting a float to an integer to display a rounded value).

`id()` and `type()` Functions:

- Display the data type and memory address of each variable using `type()` and `id()`.
- Output a summary of each variable, showing its name, value, data type, and memory location.

Program Flow:

Welcome and Instructions:

- Display a welcome message and a brief description of what the program does.

Collect Information:

- Prompt the user to enter their name (string), age (integer), height (float), and favourite number (integer).

Data Processing:

- Perform calculations with user-provided data, such as determining the birth year based on age.
- Print each variable's value, data type, and memory address using `type()` and `id()`.

Display Results:

- Print a summary of the user's information, formatted in a user-friendly way.
- Display messages showing how data types were converted where applicable.

Exit Message:

- End with a thank-you message and encourage the user to explore Python further.

Example Console Interaction:

```
Welcome to the Interactive Personal Data Collector!

Please enter your name: Alice
Please enter your age: 25
Please enter your height in meters: 1.68
Please enter your favourite number: 7

Thank you! Here is the information we collected:

Name: Alice (Type: <class 'str'>, Memory Address: 140703847239568)
Age: 25 (Type: <class 'int'>, Memory Address: 9793456)
Height: 1.68 (Type: <class 'float'>, Memory Address: 140703847253232)
Favourite Number: 7 (Type: <class 'int'>, Memory Address: 9793312)

Your birth year is approximately: 1998 (based on your age of 25)

Thank you for using the Personal Data Collector. Goodbye!
```

Instructions:

Task Completion: Ensure that you attempt all the assigned tasks given to you as part of the exam. Complete each task to the best of your ability, following the instructions provided.

Assumptions: Make suitable assumptions wherever necessary, based on the requirements and instructions provided. Document any assumptions made in your project documentation or README.md file.

GitHub Repository: Create a GitHub repository to host your project. Upload your project files, including source code, and documentation to the repository. Ensure that you provide a clear and descriptive README.md file.

No Copying: Do not copy code or any other content from your classmates or any other source. Plagiarism is strictly prohibited and can result in severe consequences, including academic penalties. Ensure that all the code and content in your project are original and properly attributed to the appropriate sources, if applicable.

Submission: Once you have completed your project and uploaded it to your GitHub repository, submit the GitHub repository link to your instructor or as instructed. Double-check that your repository is properly organized and includes all the required files as per the instructions provided.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!